Sparsh Rastogi

sparshrastogi03@gmail.com | Portfolio | LinkedIn | Google Scholar | Github

Education

Thapar Institute of Engineering and Technology — Sep 2022 – Jun 2026
Bachelor of Technology in Computer Science and Engineering — **CGPA: 9.75/10**
Relevant Coursework: Machine Learning, NLP, Generative AI, Speech Processing & Synthesis, Optimization Techniques, Probability & Statistics, Linear Algebra, Cloud Computing, Data Structures & Algorithms, Operating Systems

K.L. International School — Apr 2014 – Jul 2022
CBSE Senior Secondary Schooling (PCM with CS) — **Class 12 CBSE Score: 99.4%**

Experience

Data Science Intern | Meesho | Bengaluru, India — January 2026 – June 2026

- Built an image matching pipeline using a BEiT encoder & VLM-as-a-Judge; validated via A/B experiments (**–2.3%** duplicate listings, **+13.1%** mean group size) & scaled to production, serving ∼**4M products** daily for de-duplication and grouping.
- Curated a **1M-pair** evaluation dataset using segmentation, targeted augmentation, and hard-negative mining to benchmark the production model against **24 open-source encoders**, uncovering a **16% precision/recall** improvement opportunity and informing model selection and fine-tuning.
- Built a registry-driven knowledge base, project-specific skills, tools, and a standardized evaluation harness, enabling reliable integration of agentic AI into the development workflow, eliminating technical debt, and improving feature shipping velocity by **50%**.

Visiting Researcher | CRUISE Group, UNSW | Sydney, Australia (Remote) — October 2025 – Present

- Developed **TRACE-TS**, a framework for grounded sensor-language reasoning over wearable motion time-series, achieving **SOTA** performance across **7** benchmarks (∼**18%** macro-F1 gain over existing LLM-based methods).
- Proposed **attribution-guided reasoning distillation** to transfer expert HAR knowledge into compact time-series language models using teacher-LLM supervision and **parameter-efficient cross-modal alignment**.
- Demonstrated limitations of conventional NLG metrics for time-series reasoning and introduced **Semantic Node Match (SNM)**, a novel knowledge graph- and LLM-as-a-Judge-based evaluation metric.

Applied Scientist Intern | Amazon | Bengaluru, India — June 2025 – August 2025

- Engineered a **DML-based causal inference framework** for **110M+ users**, identifying the true causal effect of high-value actions on 365-day sales
- Optimized DML training by implementing **GPU-accelerated XGBoost on AWS SageMaker**, cutting training time by **67%**, resource usage from **16 EMR nodes to a single GPU**, and **overall cost by 8–10x.**
- Achieved **6–10% accuracy gains** using **80% less data** by optimizing surrogate windows, addressing class imbalance and validating robustness via placebo, time-series and subset tests.

Research Associate | Mediwhale | Seoul, South Korea (Remote) — February 2025 – June 2025

- Developed a multimodal system fusing **12 retinal biomarkers** and **9 wearable streams** from **1,000+ patients** to stratify risks for cardiovascular and renal diseases. [UbiComp '25]
- Validated cross-ethnic generalizability by achieving **88% accuracy** on an unseen **Brazilian cohort (16k+ patients)**, demonstrating successful transfer from Korean training data. [Circulation]
- Devised a **supervised space translation framework** to harmonize feature representations across disparate imaging devices, solving critical domain shift issues for **60 biomarkers**. [KDD-UMC '25]

Publications

- **S. Rastogi**, H. Dadwal, K. Modi, J. Bedi, and J. Singh. "Towards Sentence Level Imagined Speech Generation from EEG signals." **Interspeech**, 2025. **[Main Conference Paper]**
- **S. Rastogi**, A. Bakshi, K. Modi, S. Thakur, and V. Arora. "Integration of Wearable Health Device Data and Retinal Imaging for Multimodal Systemic Disease Stratification." **UbiComp**, 2025. **[Demo Paper]**
- S. Thakur, **S. Rastogi**, et al. "Retinal Vascular and Renal Biomarkers Predict Systemic Cardiovascular and Kidney Disease in the Brazilian Multilabel Ophthalmological Dataset (BRSET)" **Circulation (AHA)**, 2025. **[Abstract]**
- **S. Rastogi**, K. Modi, S. Thakur, and V. Arora. "Early CHD Detection from Retinal Fundus Scans Using a Spatial Context-Aware Hierarchical Attention Framework." **OMIA Workshop @ MICCAI**, 2025. **[Workshop Paper]**
- H. Dadwal, **S. Rastogi**, and J. Bedi. "Fine-Tuning Pretrained Language Models with Contextual Augmentation for Mistake Identification in Tutor–Student Dialogues." **ACL (BEA Workshop)**, 2025. **[Workshop Paper]**
- **S. Rastogi** and S. Thakur. "Enabling Cross-Device Retinal Biomarker Alignment for Generalizable Oculomics." **KDD (Undergraduate and Masters' Consortium)**, 2025. **[Workshop Paper]**

Achievements & Certifications

IndiaAI Undergraduate Fellowship | *Awarded scholarship of **INR 200,000** from Government of India* Feb 2026
AAAI Student Scholarship | *Recipient of **USD 1,840** scholarship to present research at **AAAI '26*** Nov 2025
UbiComp Travel Grant | *Awarded a grant of **EUR 1,150** to present my research at **UbiComp '25*** Oct 2025
KDD Student Travel Grant | *Received a grant of **USD 1,500** for research presentation at **KDD '25*** Aug 2025
TIET Merit Scholarship | *Earned merit scholarship worth **USD 18,000** for sustained academic excellence* May 2025
Best Oral Presentation Award | *Awarded at ICBEB '24 (Singapore) for research on retinal biomarkers* Nov 2024
JPMC Code For Good | *Winner among **53,000+ participants*** Jul 2024
Amazon ML Summer School | *Selected to attend from **20,000+ applicants*** Jul 2024
Rajasthan Police Hackathon 1.0 | *1st Runner Up among **1800+ teams** nationwide* Jan 2024

Skills

Research Interests: Multimodal Representation Learning, Causal Inference, AI for Healthcare, Large Language Models
Libraries & Frameworks: PyTorch, TensorFlow, HuggingFace, EconML, XGBoost, LangChain, OpenCV
Tools & Platforms: AWS (SageMaker, EMR, S3), PySpark, Docker, Nvidia DGX, Linux, Git
Programming Languages: Python, C/C++, SQL, Bash

Projects

NeuroSpeak | *TensorFlow, MNE, Plotly, EfficientNet, Masked Autoencoders*

- Developed a novel **Brain-Computer Interface (BCI)** for **speech synthesis** from **EEG signals**, achieving **58% accuracy** using a transformer based architecture with **masked signal modeling** and **representation learning**
- Integrating **multilingual support** and expanding to **3+ languages** by developing a **joint embedding space** using a **JEPA style architecture** for cross-lingual EEG-to-speech synthesis

Professional Service & Volunteering

- **Reviewer:** Reviewed for **MICCAI'26** conference and **workshops** at **ICML '26**, **ACL '26**, **ICLR '26**, **AAAI '26** & **ACL '25**.
- **Student Volunteer @ (AAAI' 26, UbiComp '25, KDD '25):** Assisted with conference operations and session management.
- **Director @ Studomatrix (Student NGO):** Led expansion of organisation from scratch to **50,000+ members** and secured strategic partnerships with **20+ organizations.**
- **Joint Secretary @ CCS:** Led a team of **100+** people to organize HackTU 6.0, the university's largest hackathon, with **5,000+ registrations.**